Pearson Bounds for Open-World Continual Classification

1 Classification model and regularized Pearson score

Let $P_1, \dots, P_M$ denote the distributions of the regimes represented in a fixed memory bank. An incoming batch has distribution Q. For each reference P_i, a directed empirical Pearson-type score is used to decide whether to associate the batch with that regime. A small score supports association; a large score supports keeping the incoming regime separate. An incoming regime separated from every stored reference should remain unknown rather than be merged with an existing regime.

We first analyze one fixed reference $P = P_i$. All pairwise results use this notation; Section 5.4 gives their consequence for a fixed finite bank and one common threshold.

Assumption 1.1 (Kernel and sampling)**.** Let $\mathcal{H}_k$ be a real, separable reproducing kernel Hilbert space on a measurable space $\mathcal{X}$, with measurable feature map $\phi(x) = k(\cdot, x)$. The kernel is positive semidefinite and satisfies

$$k(x, y) \geq 0, \qquad \sup_x k(x, x) \leq K, \qquad 0 < K < \infty. \tag{1}$$

Fix distributions P, Q, a positive integer n, and $\lambda > 0$. The reference observations $X_1, \dots, X_n$ are i.i.d. from P; the incoming observations $Y_1, \dots, Y_n$ are i.i.d. from Q; and all $2n$ observations are mutually independent. The regime distributions are fixed. The kernel and regularization are specified independently of the batches used to evaluate the scores.

The normalized Gaussian kernel satisfies (1) with $K = 1$. Pointwise nonnegativity is a separate requirement from positive semidefiniteness. Probabilities below integrate over the reference and incoming batches jointly; they are not conditional guarantees for every possible realized memory sample. All logarithms are natural.

1.1 Population and empirical moments

For a distribution R, define

$$\mu_R = \mathbb{E}_{Z\sim R}\phi(Z), \qquad T_R = \mathbb{E}_{Z\sim R}[\phi(Z) \otimes \phi(Z)], \tag{2}$$

$$\widehat{\mu}_P = \frac{1}{n}\sum_{r=1}^{n} \phi(X_r), \qquad \widehat{\mu}_Q = \frac{1}{n}\sum_{r=1}^{n} \phi(Y_r), \tag{3}$$

$$\widehat{T}_P = \frac{1}{n}\sum_{r=1}^{n} \phi(X_r) \otimes \phi(X_r), \qquad \Delta = \mu_Q - \mu_P, \quad \widehat{\Delta} = \widehat{\mu}_Q - \widehat{\mu}_P. \tag{4}$$

Here $(u \otimes v)w = \langle v, w\rangle_{\mathcal{H}_k} u$. The operator T_P is an uncentered second moment; subtracting $\mu_P \otimes \mu_P$ would define a different score. The bounded measurable feature map ensures that these expectations exist, as developed in Appendix A. Write

$$A = T_P + \lambda I, \qquad \widehat{A} = \widehat{T}_P + \lambda I. \tag{5}$$

For every $f \in \mathcal{H}_k$,

$$0 \le \langle f, T_P f \rangle = \mathbb{E}_P f(X)^2 \le K \|f\|^2_{\mathcal{H}_k}. \tag{6}$$

Consequently $0 \preceq T_P, \widehat{T}_P \preceq KI$ and $A, \widehat{A} \succeq \lambda I$; both regularized operators have bounded inverses.

1.2 Score and association rule

Define the population and empirical regularized Pearson-type scores

$$D_\lambda(Q;P) = \frac{1}{2}\left\langle \Delta, A^{-1}\Delta \right\rangle_{\mathcal{H}_k}, \tag{7}$$

$$\widehat{D}_\lambda = \frac{1}{2}\left\langle \widehat{\Delta}, \widehat{A}^{-1}\widehat{\Delta} \right\rangle_{\mathcal{H}_k}. \tag{8}$$

The semicolon indicates that P supplies the second-moment operator. The score is directed: interchanging P and Q generally changes it. For a positive operator B with bounded inverse, write $\|v\|^2_B = \langle v, Bv\rangle_{\mathcal{H}_k}$. Appendix G derives this regularized RKHS score from its variational definition.

For a static threshold $\eta \ge 0$, the pairwise decision is

$$\text{associate with } P \text{ if } \widehat{D}_\lambda \le \eta; \qquad \text{keep separate from } P \text{ if } \widehat{D}_\lambda > \eta. \tag{9}$$

Static means that the threshold is specified independently of the realized scores. It may depend on n, K, λ, confidence levels, and an assumed population separation. The two scenarios are

$$H_0: \quad Q = P, \tag{10}$$

$$H_1(\delta): \quad D_\lambda(Q;P) \ge \delta > 0. \tag{11}$$

The second scenario formalizes a non-merging regime relative to the reference. Distinct distributions alone do not provide a specified positive uniform margin; Appendix F explains the role of this assumption.

2 Concentration of the empirical moments

For $\gamma \in (0,1)$ define

$$b_n(\gamma) = \sqrt{\frac{2K}{n}}\left(1 + \sqrt{\log\frac{1}{\gamma}}\right), \qquad e_n(\gamma) = \frac{K}{\sqrt{n}}\left(1 + \sqrt{\log\frac{1}{\gamma}}\right). \tag{12}$$

These deterministic confidence radii are obtained using McDiarmid's inequality. The first concerns mean embeddings, the second an operator.

Lemma 2.1 (Mean-difference concentration)**.** *Under Assumption 1.1,*

$$\mathbb{P}\{\left\|\widehat{\Delta} - \Delta\right\|_{\mathcal{H}_k} \le b_n(\gamma)\} \ge 1 - \gamma. \tag{13}$$

Proof. Set $U_r = \phi(Y_r) - \mu_Q$ and $V_r = \phi(X_r) - \mu_P$. They are mutually independent and centered. "Centered" is an identity: $\mathbb{E}[\phi(X) - \mu_P] = 0$. It does not assume that $\mu_P = 0$. For a single distribution R, expansion of the squared norm gives

$$\mathbb{E}_R \|\phi(Z) - \mu_R\|^2_{\mathcal{H}_k} = \mathbb{E}_R k(Z,Z) - \|\mu_R\|^2_{\mathcal{H}_k} \le K. \tag{14}$$

Independence removes cross terms between distinct centered observations, so

$$\mathbb{E}\left\|\widehat{\Delta}-\Delta\right\|_{\mathcal{H}_k}^2 = \frac{1}{n}\left(\mathbb{E}_Q k(Y,Y) - \|\mu_Q\|_{\mathcal{H}_k}^2\right) + \frac{1}{n}\left(\mathbb{E}_P k(X,X) - \|\mu_P\|_{\mathcal{H}_k}^2\right) \leq \frac{2K}{n}. \tag{15}$$

Thus, for $F = \left\|\widehat{\Delta}-\Delta\right\|_{\mathcal{H}_k}$, $\mathbb{E}F \leq \sqrt{\mathbb{E}F^2} \leq \sqrt{2K/n}$.

Replacing one observation X_r by X_r' changes $\widehat{\Delta}$ by $-(\phi(X_r)-\phi(X_r'))/n$. The population Δ stays fixed. The reverse triangle inequality therefore gives

$$|F - F'| \leq \frac{1}{n}\left\|\phi(X_r)-\phi(X_r')\right\|_{\mathcal{H}_k},$$
$$\|\phi(x)-\phi(y)\|_{\mathcal{H}_k}^2 = k(x,x)+k(y,y)-2k(x,y) \leq 2K. \tag{16}$$

The same bound holds when replacing a Y_r. There are $2n$ observations, each with sensitivity $\sqrt{2K}/n$, and hence $\sum_{j=1}^{2n} c_j^2 = 4K/n$. McDiarmid's inequality (Appendix D) yields

$$\mathbb{P}\{F - \mathbb{E}F \geq t\} \leq \exp\left(-\frac{nt^2}{2K}\right), \qquad t > 0. \tag{17}$$

Taking $t = \sqrt{(2K/n)\log(1/\gamma)}$ proves the claim. □

Remark 2.2 (Expected error versus an individual deviation). Equation (14) bounds an expectation. It does not assert $\|\phi(X)-\mu_P\|^2 \leq K$ for every X. The coarser bound $\|\phi(X)-\mu_P\| \leq 2\sqrt{K}$ is valid pointwise and can be used to control individual summands. These two statements serve different purposes and are compatible.

Lemma 2.3 (Operator concentration)**.** *Under Assumption 1.1,*

$$\mathbb{P}\{\left\|\widehat{T}_P - T_P\right\|_{\mathrm{op}} \leq e_n(\gamma)\} \geq 1-\gamma. \tag{18}$$

Proof. The Hilbert–Schmidt operators form a Hilbert space. For $V(z) = \phi(z)\otimes\phi(z)$, the rank-one identities proved in Appendix C give

$$\|V(z)\|_{\mathrm{HS}} = k(z,z) \leq K, \qquad \langle V(x), V(y)\rangle_{\mathrm{HS}} = k(x,y)^2. \tag{19}$$

Apply the Hilbert-space variance identity to their empirical mean:

$$\mathbb{E}\left\|\widehat{T}_P - T_P\right\|_{\mathrm{HS}}^2 = \frac{\mathbb{E}_P k(X,X)^2 - \|T_P\|_{\mathrm{HS}}^2}{n} \leq \frac{K^2}{n}. \tag{20}$$

Thus $G = \left\|\widehat{T}_P - T_P\right\|_{\mathrm{HS}}$ satisfies $\mathbb{E}G \leq K/\sqrt{n}$. For one replacement,

$$|G - G'| \leq \frac{1}{n}\left\|V(x)-V(x')\right\|_{\mathrm{HS}},$$
$$\|V(x)-V(y)\|_{\mathrm{HS}}^2 = k(x,x)^2 + k(y,y)^2 - 2k(x,y)^2 \leq 2K^2. \tag{21}$$

There are n sensitivities $\sqrt{2}K/n$, so their squared sum is $2K^2/n$. McDiarmid gives

$$\mathbb{P}\{G - \mathbb{E}G \geq t\} \leq \exp(-nt^2/K^2). \tag{22}$$

Set $t = K\sqrt{\log(1/\gamma)/n}$ and use $\|B\|_{\mathrm{op}} \leq \|B\|_{\mathrm{HS}}$. □

3 Null hypothesis: association with the same regime

Theorem 3.1 (Null upper bound)**.** *Under Assumption 1.1, if $Q = P$, then for every $\alpha_0 \in (0, 1)$,*

$$\mathbb{P}\{0 \le \widehat{D}_\lambda \le U_n(\alpha_0)\} \ge 1 - \alpha_0, \tag{23}$$

where

$$\boxed{U_n(\alpha_0) = \frac{b_n(\alpha_0)^2}{2\lambda} = \frac{K}{n\lambda}\left(1 + \sqrt{\log\frac{1}{\alpha_0}}\right)^2.} \tag{24}$$

Any static threshold $\eta \ge U_n(\alpha_0)$ therefore satisfies

$$\mathbb{P}_{Q=P}\{\widehat{D}_\lambda > \eta\} \le \alpha_0. \tag{25}$$

Proof. For every realization, $\widehat{A} \succeq \lambda I$ implies $0 \preceq \widehat{A}^{-1} \preceq \lambda^{-1} I$. Hence

$$0 \le \widehat{D}_\lambda \le \frac{1}{2\lambda}\left\|\widehat{\Delta}\right\|_{\mathcal{H}_k}^2. \tag{26}$$

Under $Q = P$ the population mean difference is $\Delta = 0$. Lemma 2.1, applied at level α_0, then gives the upper bound with probability at least $1 - \alpha_0$. The event $\{\widehat{D}_\lambda > \eta\}$ is contained in $\{\widehat{D}_\lambda > U_n(\alpha_0)\}$ when $\eta \ge U_n(\alpha_0)$, proving the error statement. □

The error in (25) is failure to associate the incoming batch with its own regime. No operator concentration is needed under the null. The operator inequality is pointwise, so the dependence between $\widehat{A}$ and $\widehat{\Delta}$ is allowed.

4 Non-merging regimes: separation from a reference

Assumption 4.1 (Population separation)**.** For the fixed ordered pair and fixed regularization, $D_\lambda(Q; P) \ge \delta > 0$.

Both $\widehat{T}_P$ and $\widehat{\Delta}$ contain the reference data, so they are generally dependent. For their joint use, put $b = b_n(\alpha_1/2)$ and $e = e_n(\alpha_1/2)$ and define

$$\mathcal{E}_{b,e} = \{\left\|\widehat{\Delta} - \Delta\right\|_{\mathcal{H}_k} \le b,\ \left\|\widehat{T}_P - T_P\right\|_{\mathrm{op}} \le e\}. \tag{27}$$

The two concentration lemmas and a union bound give

$$\mathbb{P}(\mathcal{E}_{b,e}) \ge 1 - \alpha_1. \tag{28}$$

Independence between these two error events is not required.

Lemma 4.2 (Inverse-operator comparison)**.** *If $\left\|\widehat{T}_P - T_P\right\|_{\mathrm{op}} \le e$ and $c = 1 + e/\lambda$, then*

$$\widehat{A}^{-1} \succeq c^{-1} A^{-1}. \tag{29}$$

Proof. Self-adjointness implies $\widehat{T}_P - T_P \preceq eI$. Since $A \succeq \lambda I$,

$$\widehat{A} \preceq A + eI \preceq (1 + e/\lambda)A = cA.$$

Inversion reverses the order of positive operators with bounded inverses, proving the claim. Appendix E justifies this step without assuming commutativity. □

Theorem 4.3 (Non-merging lower bound)**.** *Under Assumptions 1.1 and 4.1, for every* $\alpha_1 \in (0,1)$*,*

$$\mathbb{P}\{\widehat{D}_\lambda \geq L_n(\delta, \alpha_1)\} \geq 1 - \alpha_1, \tag{30}$$

where

$$\boxed{L_n(\delta, \alpha_1) = \frac{\left[\sqrt{2\delta} - b_n(\alpha_1/2)/\sqrt{\lambda}\right]_+^2}{2\left(1 + e_n(\alpha_1/2)/\lambda\right)}.} \tag{31}$$

Here $[x]_+ = \max\{x, 0\}$*. Any* $\eta < L_n(\delta, \alpha_1)$ *satisfies*

$$\mathbb{P}\{\widehat{D}_\lambda \leq \eta\} \leq \alpha_1 \quad \textit{for each fixed pair with } D_\lambda(Q;P) \geq \delta. \tag{32}$$

Proof. Use b, e from (27), put $c = 1 + e/\lambda$, and work on $\mathcal{E}_{b,e}$. The inverse comparison gives

$$\sqrt{2\widehat{D}_\lambda} = \left\|\widehat{\Delta}\right\|_{\widehat{A}^{-1}} \geq \frac{1}{\sqrt{c}} \left\|\widehat{\Delta}\right\|_{A^{-1}}. \tag{33}$$

Operator order holds for every vector, so it applies to the realized $\widehat{\Delta}$ even though it depends on $\widehat{A}$. The reverse triangle inequality in the norm induced by A^{-1} gives

$$\left\|\widehat{\Delta}\right\|_{A^{-1}} \geq \left[\|\Delta\|_{A^{-1}} - \left\|\widehat{\Delta} - \Delta\right\|_{A^{-1}}\right]_+.$$

Since $A^{-1} \preceq \lambda^{-1} I$,

$$\left\|\widehat{\Delta} - \Delta\right\|_{A^{-1}} \leq \frac{1}{\sqrt{\lambda}} \left\|\widehat{\Delta} - \Delta\right\|_{\mathcal{H}_k} \leq \frac{b}{\sqrt{\lambda}}.$$

Moreover $\|\Delta\|_{A^{-1}} = \sqrt{2D_\lambda(Q;P)} \geq \sqrt{2\delta}$. Combining these facts gives

$$\sqrt{2\widehat{D}_\lambda} \geq \frac{1}{\sqrt{c}} \left[\sqrt{2\delta} - b/\sqrt{\lambda}\right]_+.$$

Both sides are nonnegative, so squaring proves the lower bound. Its event has probability at least $1 - \alpha_1$ by the union bound. If $\eta < L_n$, association implies $\widehat{D}_\lambda < L_n$, giving the error bound. □

The error in (32) is an incorrect merge with a separated reference. The mean error reduces the usable separation by $b/\sqrt{\lambda}$, and the operator error introduces the factor $1 + e/\lambda$. The bound is valid for every $e \geq 0$; its lower endpoint is positive only if $\sqrt{2\delta} > b/\sqrt{\lambda}$.

5 A nonempty interval for a static classification threshold

Associate with the reference when $\widehat{D}_\lambda \leq \eta$, and keep it separate when $\widehat{D}_\lambda > \eta$. Under $Q = P$, keeping the batch separate is an error. Under a non-merging regime, association with the reference is an incorrect merge. These are different probability models, so their error levels are not added to calibrate the two tests.

Theorem 5.1 (Admissible static-threshold interval)**.** *For given* $\alpha_0, \alpha_1 \in (0,1)$*, any deterministic threshold with*

$$U_n(\alpha_0) \leq \eta < L_n(\delta, \alpha_1) \tag{34}$$

satisfies

$$\mathbb{P}_{Q=P}\{\widehat{D}_\lambda > \eta\} \leq \alpha_0, \qquad \mathbb{P}_{D_\lambda(Q;P) \geq \delta}\{\widehat{D}_\lambda \leq \eta\} \leq \alpha_1. \tag{35}$$

The second notation means each fixed pair satisfying the separation assumption; it is not a probability distribution on pairs.

Proof. The first error event is contained in $\{\widehat{D}_\lambda > U_n(\alpha_0)\}$. Because $\eta < L_n$, the second is contained in $\{\widehat{D}_\lambda < L_n(\delta, \alpha_1)\}$. Apply the two theorems. The strict inequality at the right endpoint accounts for the acceptance rule at equality; it cannot in general be replaced by $\eta \le L_n$. □

Proposition 5.2 (Nonempty threshold interval)**.** *Put*

$$b_0 = b_n(\alpha_0), \quad b_1 = b_n(\alpha_1/2), \quad c = 1 + e_n(\alpha_1/2)/\lambda.$$

Since $b_0 > 0$, the interval (34) *is nonempty if and only if*

$$\boxed{\delta > \frac{(b_1 + \sqrt{c}\, b_0)^2}{2\lambda}.} \tag{36}$$

Proof. The inequality $L_n > U_n$ is equivalent to $\left[\sqrt{2\delta} - b_1/\sqrt{\lambda}\right]_+ > \sqrt{c}\, b_0/\sqrt{\lambda}$. The right side is positive, so the positive part can be removed and the resulting positive inequality can be squared. □

5.1 Common tolerance and explicit calibration

For $\alpha_0 = \alpha_1 = \alpha$, choose

$$\boxed{\eta = \frac{K}{n\lambda}\left(1 + \sqrt{\log\frac{1}{\alpha}}\right)^2.} \tag{37}$$

It controls incorrect separation under the null at level α without a population separation assumption. The sufficient separation for the non-merging guarantee is

$$\begin{aligned}\delta > \frac{K}{n\lambda}\Bigg[1 + \sqrt{\log\frac{2}{\alpha}} \\ + \left(1 + \sqrt{\log\frac{1}{\alpha}}\right)\sqrt{1 + \frac{K}{\lambda\sqrt{n}}\left(1 + \sqrt{\log\frac{2}{\alpha}}\right)}\Bigg]^2.\end{aligned} \tag{38}$$

This is equivalent to nonemptiness of the displayed interval. It is not necessary for actual successful classification: the concentration radii may overestimate the realized errors.

5.2 Sample size and regularization

For fixed K, λ, α and fixed separated pair, $b_n, e_n \to 0$, $U_n \to 0$, and $L_n(\delta, \alpha) \to \delta$. Thus the certified interval eventually becomes nonempty. For fixed confidence, the two quantities controlling usefulness are

$$\frac{b_n}{\sqrt{\lambda}} = O((n\lambda)^{-1/2}), \qquad \frac{e_n}{\lambda} = O((\lambda\sqrt{n})^{-1}).$$

These expressions explain why very small regularization makes the uniform bounds demanding. If $\lambda = \lambda_n$ changes, the population score changes, so the validity and size of an assumed separation must also be checked. One cannot optimize the explicit threshold while treating separation as a fixed unrelated constant.

5.3 An explicit sufficient batch size

Corollary 5.3 (Batch size ensuring a nonempty interval)**.** *For common tolerance α, put*

$$a_0 = 1 + \sqrt{\log(1/\alpha)}, \qquad a_1 = 1 + \sqrt{\log(2/\alpha)}.$$

The interval is nonempty if the integer n satisfies both

$$n \geq \left(\frac{K a_1}{\lambda}\right)^2, \qquad n > \frac{K(a_1 + \sqrt{2}\, a_0)^2}{\lambda \delta}. \tag{39}$$

Proof. The first condition gives $e_n(\alpha/2)/\lambda \leq 1$, so $c \leq 2$. Since $b_0 = \sqrt{2K/n}\, a_0$ and $b_1 = \sqrt{2K/n}\, a_1$, the right side of (36) is at most $K(a_1 + \sqrt{2}\, a_0)^2/(n\lambda)$. The second condition makes this strictly smaller than δ. □

5.4 One static threshold for a fixed memory bank

Let $P_1, \ldots, P_M$ be distinct fixed reference regimes and $\mathcal{Q}$ an admissible family of incoming distributions. Use the same n, K, λ for all comparisons. Assume a uniform non-merging margin $\delta_* > 0$ such that

$$D_\lambda(Q; P_i) \geq \delta_* \quad \text{for every } Q \in \mathcal{Q} \text{ and every } i \text{ with } Q \neq P_i. \tag{40}$$

This includes all wrong-reference comparisons for known regimes and all reference comparisons for novel regimes. Separation within the stored bank alone would not ensure separation for arbitrary unknown distributions.

Corollary 5.4 (Known and unknown regime classification)**.** *Suppose each reference comparison satisfies Assumption 1.1 and* (40)*. Scores may share the incoming batch; reference batches need not be independent of one another. For $\alpha \in (0, 1)$, choose a common static threshold with*

$$U_n(\alpha/M) \leq \eta < L_n(\delta_*, \alpha/M). \tag{41}$$

Let $\widehat{\mathcal{A}}(Q) = \{i : \widehat{D}_{\lambda,i} \leq \eta\}$. For $Q = P_j \in \mathcal{Q}$,

$$\mathbb{P}\{\widehat{\mathcal{A}}(Q) = \{j\}\} \geq 1 - \alpha. \tag{42}$$

For $Q \in \mathcal{Q}$ different from every stored regime,

$$\mathbb{P}\{\widehat{\mathcal{A}}(Q) = \varnothing\} \geq 1 - \alpha. \tag{43}$$

A common threshold exists whenever (38) *holds with $\delta = \delta_*$ and α replaced by α/M.*

Proof. If $Q = P_j$, the null theorem bounds the event $\widehat{D}_{\lambda,j} > \eta$ by α/M. For every $i \neq j$, the non-merging theorem bounds $\widehat{D}_{\lambda,i} \leq \eta$ by α/M. A union bound over these M undesired decisions gives failure probability at most α; on their complement exactly j is accepted. For a novel Q, all M reference comparisons satisfy the non-merging hypothesis. A union bound on their association events gives an empty accepted set with probability at least $1 - \alpha$. No independence between scores is used. Nonemptiness follows from (36). □

The corresponding decision associates a batch when exactly one reference is accepted, retains it as unknown when none are accepted, and abstains when multiple references are accepted. The guarantee is per incoming batch for a fixed bank under the stated sampling model. It does not assume that an arbitrarily selected or adaptively constructed memory sample has those same sampling properties.

A RKHS and expectation foundations

A.1 The reproducing property

For $\phi(x) = k(\cdot, x)$,

$$f(x) = \langle f, \phi(x) \rangle_{\mathcal{H}_k}, \qquad \langle \phi(x), \phi(y) \rangle_{\mathcal{H}_k} = k(x, y), \qquad \|\phi(x)\|^2_{\mathcal{H}_k} = k(x, x).$$

Cauchy–Schwarz implies $|k(x, y)| \le \sqrt{k(x, x)k(y, y)} \le K$ and $|f(x)| \le \sqrt{K}\, \|f\|_{\mathcal{H}_k}$. A Hilbert-valued random element has a Bochner expectation when it is strongly measurable and its expected norm is finite. Our assumptions give these properties, since $\|\phi(X)\| \le \sqrt{K}$ and $\|\phi(X) \otimes \phi(X)\|_{\mathrm{HS}} \le K$. Linear functionals pass through these expectations, which justifies identities such as $\mathbb{E} f(X) = \langle f, \mu_P \rangle$.

A.2 Boundedness and positivity of the second moment

For $f \in \mathcal{H}_k$,

$$0 \le \langle f, T_P f \rangle = \mathbb{E}_P f(X)^2 \le K \|f\|^2_{\mathcal{H}_k}.$$

This proves $0 \preceq T_P \preceq KI$. The empirical version is proved by replacing the expectation with a finite average. Consequently A^{-1} and $\widehat{A}^{-1}$ exist, and each has operator norm at most $1/\lambda$.

A.3 Why the score can be zero for distinct distributions

Since $A \preceq (K + \lambda)I$,

$$\frac{\|\Delta\|^2}{2(K + \lambda)} \le D_\lambda(Q; P) \le \frac{\|\Delta\|^2}{2\lambda}. \tag{44}$$

Thus zero score is equivalent to equal mean embeddings. A noncharacteristic kernel can map different distributions to the same mean embedding. Positive score separation must therefore be assumed or established using properties of the kernel and the admissible distributions.

B Centering, variance, and independence

B.1 The Hilbert-space variance identity

Let W be a square-integrable random element of a real Hilbert space and $\mu = \mathbb{E}W$. Expand:

$$\begin{aligned} \mathbb{E}\|W - \mu\|^2 &= \mathbb{E}\|W\|^2 - 2\mathbb{E}\langle W, \mu \rangle + \|\mu\|^2 \\ &= \mathbb{E}\|W\|^2 - 2\langle \mathbb{E}W, \mu \rangle + \|\mu\|^2 \\ &= \mathbb{E}\|W\|^2 - \|\mu\|^2. \end{aligned}$$

The middle term does not vanish: it equals $-2\|\mu\|^2$. It combines with the last term to leave a subtraction. No assumption that the uncentered observation has mean zero is used.

B.2 Independent empirical averages

If $W_1, \ldots, W_n$ are independent copies of W, let $R_r = W_r - \mu$. Independence gives, for $r \ne s$,

$$\mathbb{E}\langle R_r, R_s \rangle = \langle \mathbb{E}R_r, \mathbb{E}R_s \rangle = 0.$$

Hence

$$\mathbb{E}\left\| \frac{1}{n} \sum_r R_r \right\|^2 = \frac{1}{n^2} \sum_{r,s} \mathbb{E}\langle R_r, R_s \rangle = \frac{1}{n} \left(\mathbb{E}\|W\|^2 - \|\mu\|^2 \right).$$

B.3 An example separating a pointwise bound from an expectation

Consider two points a, b with $k(x, y) = \mathbf{1}\{x = y\}$ and $P(a) = p$, $P(b) = 1 - p$. The feature vectors are orthonormal, $K = 1$, and

$$\|\phi(a) - \mu_P\|^2 = 2(1-p)^2, \qquad \mathbb{E}\,\|\phi(X) - \mu_P\|^2 = 2p(1-p) \le \frac{1}{2}.$$

At $p = 0.01$, the first quantity is $1.9602 > K$, but its expectation is $0.0198 < K$. An individual large error does not contradict a small second moment. In particular, the exact second-moment estimate cannot be substituted for a pointwise summand bound in a concentration theorem.

B.4 What dependence would change

If $U = \widehat{\mu}_Q - \mu_Q$ and $V = \widehat{\mu}_P - \mu_P$, then always

$$\mathbb{E}\,\|U - V\|^2 = \mathbb{E}\,\|U\|^2 + \mathbb{E}\,\|V\|^2 - 2\mathbb{E}\,\langle U, V\rangle\,.$$

Centering alone does not remove the last term. Independent windows do remove it. Dependence within a window likewise changes the variance of that window's mean. The classification results require the independence assumptions in Assumption 1.1.

C Rank-one operators and Hilbert–Schmidt norms

Let (e_j) be an orthonormal basis of $\mathcal{H}_k$. A Hilbert–Schmidt operator B has $\|B\|_{\mathrm{HS}}^2 = \sum_j \|Be_j\|^2$. For rank-one operators, Parseval's identity gives

$$\|u \otimes v\|_{\mathrm{HS}}^2 = \sum_j \|\langle v, e_j\rangle\, u\|^2 = \|u\|^2 \sum_j |\,\langle v, e_j\rangle\,|^2 = \|u\|^2\,\|v\|^2\,,$$
$$\langle u \otimes v, a \otimes b\rangle_{\mathrm{HS}} = \langle u, a\rangle\,\langle v, b\rangle\,.$$

Taking $u = v = \phi(x)$ and $a = b = \phi(y)$ gives (19). Expanding the squared difference then gives (21). For any unit vector f, extending f to an orthonormal basis shows $\|Bf\|^2 \le \|B\|_{\mathrm{HS}}^2$; taking a supremum proves $\|B\|_{\mathrm{op}} \le \|B\|_{\mathrm{HS}}$.

It is the error $\widehat{T}_P - T_P$ that is Hilbert–Schmidt. In an infinite-dimensional RKHS, I need not be Hilbert–Schmidt, and we do not claim that $A = T_P + \lambda I$ is Hilbert–Schmidt. We use operator norms and order for the regularized operators instead.

D McDiarmid's inequality and constant bookkeeping

Let $Z_1, \ldots, Z_N$ be independent, and let f be a measurable real function such that changing coordinate j changes f by at most a deterministic c_j. Then

$$\mathbb{P}\{f - \mathbb{E}f \ge t\} \le \exp\left(-\frac{2t^2}{\sum_{j=1}^N c_j^2}\right). \tag{45}$$

It follows that, with probability at least $1 - \gamma$,

$$f \le \mathbb{E}f + \sqrt{\frac{1}{2}\sum_j c_j^2 \log(1/\gamma)}.$$

The squared sum must count *all* independent input coordinates. For the mean difference there are $2n$; for the reference operator there are n.

Function	Inputs	Sensitivity	Squared sum
$\left\|\widehat{\Delta}-\Delta\right\|$	$2n$	$\sqrt{2K}/n$	$4K/n$
$\left\|\widehat{T}_P-T_P\right\|_{\mathrm{HS}}$	n	$\sqrt{2}K/n$	$2K^2/n$

For a two-sided scalar deviation around the expectation, a union bound over the upper and lower tails introduces a factor 2 in probability, or $\log(2/\gamma)$ in the radius. Our main lemmas need only an upper tail for an already nonnegative error norm, so they use $\log(1/\gamma)$ instead.

E Operator order and the weighted norm

E.1 Order and inversion

For self-adjoint operators, $B \preceq C$ means $\langle v, Bv\rangle \le \langle v, Cv\rangle$ for every v. If $0 \prec B \preceq C$ with bounded inverses, let $S = C^{-1/2}BC^{-1/2}$. Then $0 \prec S \preceq I$, so its spectral values are in $(0,1]$ and $S^{-1} \succeq I$. Consequently,

$$B^{-1} = C^{-1/2}S^{-1}C^{-1/2} \succeq C^{-1}.$$

This proves order reversal without multiplying noncommuting inequalities in an invalid order. Apply this to $B = \widehat{A}$ and $C = cA$ in Lemma 4.2.

E.2 The reverse triangle inequality in the weighted norm

Since A^{-1} is positive with bounded inverse,

$$\|v\|_{A^{-1}} = \left\|A^{-1/2}v\right\|_{\mathcal{H}_k}$$

is a norm. Its triangle inequality is the Hilbert-space triangle inequality after applying the linear map $A^{-1/2}$. Therefore

$$\|u\|_{A^{-1}} \ge \|v\|_{A^{-1}} - \|u-v\|_{A^{-1}}.$$

The left side is nonnegative, so the positive part can be taken on the right. This is the positive part in the non-merging bound. Also $A^{-1} \preceq \lambda^{-1}I$ implies $\|u-v\|_{A^{-1}} \le \|u-v\|_{\mathcal{H}_k}/\sqrt{\lambda}$.

E.3 Why shared reference observations are allowed

For any two events, a union bound gives $\mathbb{P}(E_1 \cap E_2) \ge 1 - \mathbb{P}(E_1^c) - \mathbb{P}(E_2^c)$. Once both error bounds hold, operator order is an inequality for every vector on that outcome. It can be applied to the realized $\widehat{\Delta}$ without assuming that it is independent of $\widehat{A}$. In particular, the argument does not factor an expectation involving these estimators.

E.4 An explicit example where the estimators are coupled

Take $\mathcal{X} = \{0,1\}$ and the scalar kernel $k(x,y) = xy$. It is positive semidefinite and nonnegative, with $K = 1$; identify $\phi(x) = x$ in a one-dimensional feature space. Let $n = 1$, $P = Q =$ Bernoulli(0.2), X, Y independent, and $\lambda = 1$. Then

$$\widehat{T}_P = X, \quad \widehat{\Delta} = Y - X, \quad \widehat{A}^{-1} = \frac{1}{1+X}, \quad \widehat{D}_\lambda = \frac{(Y-X)^2}{2(1+X)}.$$

The events $(X,Y) = (0,1)$ and $(1,0)$ each have probability 0.16. Their scores are $1/2$ and $1/4$, respectively; the other scores vanish. Therefore

$$\mathbb{E}\widehat{D}_\lambda = 0.16\left(\frac{1}{2}+\frac{1}{4}\right) = 0.12, \qquad \frac{1}{2}\mathbb{E}[\widehat{\Delta}^2]\mathbb{E}[\widehat{A}^{-1}] = \frac{1}{2}(0.32)(0.9) = 0.144.$$

The failed factorization demonstrates actual dependence. Our null domination and joint-event operator argument remain valid because neither uses this factorization.

F Separation, identifiability, and sufficient conditions

For a fixed finite collection with injective mean embeddings and fixed λ, every distinct ordered pair has positive score. The minimum over finitely many such positive scores is positive. This supports a finite-bank separation assumption. It does not imply a positive minimum over all distinct distributions. To see why, fix P, R and let $Q_\varepsilon = (1-\varepsilon)P + \varepsilon R$. Then

$$\mu_{Q_\varepsilon} - \mu_P = \varepsilon(\mu_R - \mu_P), \qquad D_\lambda(Q_\varepsilon; P) = \varepsilon^2 D_\lambda(R; P) \longrightarrow 0.$$

This example applies even to characteristic Gaussian kernels. A uniform positive separation for unknown regimes therefore restricts the allowed alternatives; it is not a consequence of $P \neq Q$.

The static-threshold separation criterion (36) is necessary and sufficient for nonemptiness of the interval defined by our chosen U_n, L_n. It is only a sufficient condition for small actual error. Failure of this criterion means the displayed bounds do not certify the desired threshold, not that classification is impossible.

G Variational derivation of the classification score

For a reference distribution P, the regularized RKHS Pearson functional is

$$\sup_{f \in \mathcal{H}_k} \left\{ \mathbb{E}_Q f - \mathbb{E}_P f - \frac{1}{2}\mathbb{E}_P f^2 - \frac{\lambda}{2}\|f\|^2_{\mathcal{H}_k} \right\}. \tag{46}$$

The reproducing property identifies the expression in braces with $\langle f, \Delta\rangle - \frac{1}{2}\langle f, Af\rangle$. Completing the square gives

$$\langle f, \Delta\rangle - \frac{1}{2}\langle f, Af\rangle = \frac{1}{2}\left\langle \Delta, A^{-1}\Delta \right\rangle - \frac{1}{2}\left\| f - A^{-1}\Delta \right\|^2_A.$$

The final squared norm is nonnegative and vanishes only at $f = A^{-1}\Delta$. The supremum is therefore exactly (7). Replacing population expectations by empirical averages yields (8), with maximizer $\widehat{A}^{-1}\widehat{\Delta}$. The factor $1/2$ is part of the normalization, and the restriction to $\mathcal{H}_k$ and the regularization are part of the score itself. The separation assumption refers to this score, not to an unrestricted density-ratio functional.

H Expansion of the static-threshold condition

At common tolerance α, write

$$a_0 = 1 + \sqrt{\log(1/\alpha)}, \qquad a_1 = 1 + \sqrt{\log(2/\alpha)}.$$

Then

$$b_0 = \sqrt{\frac{2K}{n}}a_0, \qquad b_1 = \sqrt{\frac{2K}{n}}a_1, \qquad c = 1 + \frac{K}{\lambda\sqrt{n}}a_1.$$

Substituting in (36),

$$\begin{aligned}\frac{(b_1 + \sqrt{c}\, b_0)^2}{2\lambda} &= \frac{1}{2\lambda}\frac{2K}{n}\left(a_1 + \sqrt{1 + \frac{K}{\lambda\sqrt{n}}a_1}\, a_0\right)^2 \\ &= \frac{K}{n\lambda}\left(a_1 + a_0\sqrt{1 + \frac{K}{\lambda\sqrt{n}}a_1}\right)^2.\end{aligned}$$

Expanding a_0, a_1 gives (38). The strict inequality means that the lower confidence bound for a non-merging regime lies strictly above the null upper confidence bound. These are bounds for two different population scenarios. Their separation certifies that one predetermined threshold can implement both classification decisions at the stated error levels.